

# A Practical Guide to Overfitting machine learning

---

TOPIC -- OVERFITTING MACHINE LEARNING

-- 01 --

Overfitted models ... are often free of bias in the parameter estimators, but have estimated (and actual) sampling variances that are needlessly large (the precision of the estimators is poor, relative to what could have been accomplished with a more parsimonious model). False treatment effects tend to be identified, and false variables are included with overfitted models. ... A best approximating model is achieved by properly balancing the errors of underfitting and overfitting. Overfitting is more likely to be a serious concern when there is little theory available to guide the analysis, in part because then there tend to be a large number of models to select from. The book *Model Selection and Model Averaging* (2008) puts it this way.

-- 02 --

Regression In regression analysis, overfitting occurs frequently. As an extreme example, if there are  $p$  variables in a linear regression with  $p$  data points, the fitted line can go exactly through every point. For logistic regression or Cox proportional hazards models, there are a variety of rules of thumb (e.g. 5–9, 10 and 10–15 — the guideline of 10 observations per independent variable is known as the "one in ten rule"). In the process of regression model selection, the mean squared error of the random regression function can be split into random noise, approximation bias, and variance in the estimate of the regression function. The bias–variance tradeoff is often used to overcome overfit models. With a large set of explanatory variables that actually have no relation to the dependent variable being predicted, some variables will in general be falsely found to be statistically significant and the researcher may thus retain them in the model, thereby overfitting the model. This is known as Freedman's paradox.

-- 03 --

Usually, a learning algorithm is trained using some set of "training data": exemplary situations for which the desired output is known. The goal is that the algorithm will also perform well on predicting the output when fed "validation data" that was not encountered during its training. Overfitting is the use of models or procedures that violate Occam's razor, for example by including more adjustable parameters than are ultimately optimal, or by using a more complicated approach than is ultimately optimal. For an example where there are

too many adjustable parameters, consider a dataset where training data for  $y$  can be adequately predicted by a linear function of two independent variables. Such a function requires only three parameters (the intercept and two slopes). Replacing this simple function with a new, more complex quadratic function, or with a new, more complex linear function on more than two independent variables, carries a risk: Occam's razor implies that any given complex function is a priori less probable than any given simple function. If the new, more complicated function is selected instead of the simple function, and if there was not a large enough gain in training data fit to offset the complexity increase, then the new complex function "overfits" the data and the complex overfitted function will likely perform worse than the simpler function on validation data outside the training dataset, even though the complex function performed as well, or perhaps even better, on the training dataset. When comparing different types of models, complexity cannot be measured solely by counting how many parameters exist in each model; the expressivity of each parameter must be considered as well. For example, it is nontrivial to directly compare the complexity of a neural net (which can track curvilinear relationships) with  $m$  parameters to a regression model with  $n$  parameters. Overfitting is especially likely in cases where learning was performed too long or where training examples are rare, causing the learner to adjust to very specific random features of the training data that have no causal relation to the target function. In this process of overfitting, the performance on the training examples still increases while the performance on unseen data becomes worse. As a simple example, consider a database of retail purchases that includes the item bought, the purchaser, and the date and time of purchase. It's easy to construct a model that will fit the training set perfectly by using the date and time of purchase to predict the other attributes, but this model will not generalize at all to new data because those past times will never occur again. Generally, a learning algorithm is said to overfit relative to a simpler one if it is more accurate in fitting known data (hindsight) but less accurate in predicting new data (foresight). One can intuitively understand overfitting from the fact that information from all past experience can be divided into two groups: information that is relevant for the future, and irrelevant information ("noise"). Everything else being equal, the more difficult a criterion is to predict (i.e., the higher its uncertainty), the more noise exists in past information that needs to be ignored. The problem is determining which part to ignore. A learning algorithm that can reduce the risk of fitting noise is called "robust."

-- 04 --

A function that is overfitted is likely to request more information about each item in the validation dataset than does the optimal function; gathering this additional

unnneeded data can be expensive or error-prone, especially if each individual piece of information must be gathered by human observation and manual data entry. A more complex, overfitted function is likely to be less portable than a simple one. At one extreme, a one-variable linear regression is so portable that, if necessary, it could even be done by hand. At the other extreme are models that can be reproduced only by exactly duplicating the original modeler's entire setup, making reuse or scientific reproduction difficult. It may be possible to reconstruct details of individual training instances from an overfitted machine learning model's training set. This may be undesirable if, for example, the training data includes sensitive personally identifiable information (PII). This phenomenon also presents problems in the area of artificial intelligence and copyright, with the developers of some generative deep learning models such as Stable Diffusion and GitHub Copilot being sued for copyright infringement because these models have been found to be capable of reproducing certain copyrighted items from their training data.